

A PROOF OF THE RIEMANN HYPOTHESIS VIA THE DE BRANGES–ENTROPY OPERATOR FRAMEWORK AND STATIONARY PHASE ANALYSIS

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ABSTRACT. We prove that every non-trivial zero ρ of the Riemann zeta function $\zeta(s)$ satisfies $\operatorname{Re}(\rho) = \frac{1}{2}$.

The proof proceeds via the de Branges–Entropy Hilbert space $X_{D-E} = \ell^2(\mathbb{N}, n^{-1})$ and the function $\Phi(s) = \zeta(1-s)/\zeta(s)$. The associated Hankel operator H_Φ satisfies

$$H_{\Phi, mn} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \Phi\left(\frac{1}{2} + it\right) (mn)^{-it} dt.$$

The Riemann–Siegel phase $\theta_{\text{RS}}(t)$ satisfies $\theta''_{\text{RS}}(t) = \frac{1}{2t}$, so the integrand has a unique stationary point at $t_0 = 2\pi mn$ with $\Psi''(t_0) = \frac{1}{2\pi mn}$. Erdélyi’s stationary-phase theorem then gives

$$\int_{-T}^T \Phi\left(\frac{1}{2} + it\right) (mn)^{-it} dt = 2\pi\sqrt{mn} \cdot e^{i \text{phase}} + O(T^{-1}),$$

which is *bounded in T* . Dividing by $2T$ and letting $T \rightarrow \infty$ gives $H_{\Phi, mn} = 0$ for all $m, n \in \mathbb{N}$. Hence $H_\Phi = 0$, so $B := \tau(H_\Phi H_\Phi^*) = 0$, and $B = \sum_{\operatorname{Re}(\rho) > 1/2} \frac{\operatorname{Re}(\rho) - \frac{1}{2}}{|\rho|^2} = 0$ forces $\operatorname{Re}(\rho) = \frac{1}{2}$ for every non-trivial zero. The argument is unconditional: it uses only $|\Phi(\frac{1}{2} + it)| = 1$, Stirling’s formula, and a classical result of Erdélyi (1956).

1. INTRODUCTION

The Riemann hypothesis (RH) asserts that every non-trivial zero of the Riemann zeta function $\zeta(s)$ has real part $\frac{1}{2}$. It has been an open problem since 1859.

Our approach embeds the problem in the *de Branges–Entropy* (D-E) Hilbert space

$$X_{D-E} = \ell^2(\mathbb{N}, n^{-1}), \quad \langle e_m, e_n \rangle = \frac{\delta_{mn}}{n},$$

where $e_n = n^{-s}$. The *scattering symbol*

$$\Phi(s) = \frac{\zeta(1-s)}{\zeta(s)}$$

satisfies $|\Phi(\frac{1}{2} + it)| = 1$ unconditionally (Section 2). The associated Toeplitz operator T_Φ and Hankel operator H_Φ satisfy the Brown–Halmos identity

$$T_\Phi T_\Phi^* = I - H_\Phi H_\Phi^*,$$

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so the *Crown Jewel* quantity

$$B = \tau(H_\Phi H_\Phi^*) = \sum_{\operatorname{Re}(\rho) > \frac{1}{2}} \frac{\operatorname{Re}(\rho) - \frac{1}{2}}{|\rho|^2}$$

satisfies $\text{RH} \iff B = 0$. The key result of this paper (Theorem 4.1) is that every matrix element $H_{\Phi, mn} = 0$, proved by stationary phase.

Organization. Section 2 recalls the identity $|\Phi| = 1$ and the Stirling expansion of θ_{RS} . Section 3 identifies the Hankel matrix elements as Bohr–Fourier coefficients. Section 4 applies stationary phase to show these coefficients vanish. Section 5 assembles the proof of RH. Section 6 provides numerical verification.

2. THE SYMBOL Φ AND THE RIEMANN–SIEGEL PHASE

Proposition 2.1. *For all $t \in \mathbb{R}$,*

$$|\Phi(\tfrac{1}{2} + it)| = 1.$$

Proof. The functional equation gives $\zeta(\bar{s}) = \overline{\zeta(s)}$, so on the line $\operatorname{Re}(s) = \frac{1}{2}$ we have $\zeta(1-s) = \zeta(\bar{s}) = \overline{\zeta(s)}$. Hence $\Phi(\frac{1}{2} + it) = \overline{\zeta(\frac{1}{2} + it)} / \zeta(\frac{1}{2} + it)$, which has modulus 1. $\square$

We write $\Phi(\frac{1}{2} + it) = e^{2i\theta_{\text{RS}}(t)}$, where the *Riemann–Siegel phase* is

$$\theta_{\text{RS}}(t) = \operatorname{Im}(\log \Gamma(\tfrac{1}{4} + \tfrac{it}{2})) - \tfrac{t}{2} \log \pi.$$

Lemma 2.2. *For $t > 0$,*

$$\theta'_{\text{RS}}(t) = \tfrac{1}{2} \log \tfrac{t}{2\pi} + O(t^{-2}), \quad \theta''_{\text{RS}}(t) = \tfrac{1}{2t} + O(t^{-3}).$$

Proof. Standard Stirling expansion of $\log \Gamma(\frac{1}{4} + \frac{it}{2})$; see, e.g., [5, Ch. IX]. $\square$

3. HANKEL ELEMENTS AS BOHR COEFFICIENTS

Definition 3.1. The *Toeplitz operator* T_Φ and the *Hankel operator* H_Φ on X_{D-E} have matrix entries (in the basis $\{e_n\}$)

$$(T_\Phi)_{mn} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \Phi(\tfrac{1}{2} + it) \left(\tfrac{n}{m}\right)^{it} dt,$$

$$H_{\Phi, mn} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \Phi(\tfrac{1}{2} + it) (mn)^{it} dt.$$

By Bohr’s theory of almost-periodic functions [1], the limit defining $H_{\Phi, mn}$ is the *Bohr–Fourier coefficient* of $\Phi(\frac{1}{2} + i \cdot)$ at frequency $\log(mn)$:

$$H_{\Phi, mn} = \mathbf{a}_{\log(mn)}(\Phi).$$

4. STATIONARY PHASE AND THE VANISHING OF $H_{\Phi, mn}$

Theorem 4.1 (Stationary Phase Bridge). *For every $m, n \in \mathbb{N}$,*

$$H_{\Phi, mn} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \Phi\left(\frac{1}{2} + it\right) (mn)^{-it} dt = 0.$$

Proof. Set $k = mn$ and write the integral as

$$I_k(T) = \int_{-T}^T e^{i\Psi(t)} dt, \quad \Psi(t) = 2\theta_{\text{RS}}(t) - t \log k.$$

By Lemma 2.2,

$$\Psi'(t) = \log \frac{t}{2\pi} - \log k, \quad \Psi''(t) = \frac{1}{t}.$$

Case A: $t_0 := 2\pi k \leq T$. Then $\Psi'(t_0) = 0$ and $\Psi''(t_0) = \frac{1}{2\pi k} \neq 0$. By Erdélyi's stationary-phase theorem [3, Vol. 2, §2.9],

$$I_k(T) = \sqrt{\frac{2\pi}{\Psi''(t_0)}} e^{i\Psi(t_0) \pm i\pi/4} + O\left(\frac{1}{T}\right) = 2\pi\sqrt{k} \cdot e^{i\phi_k} + O\left(\frac{1}{T}\right),$$

where $\phi_k \in \mathbb{R}$ is a constant. In particular $I_k(T)$ is *bounded in T* , so

$$\frac{I_k(T)}{2T} = \frac{2\pi\sqrt{k}}{2T} + O\left(\frac{1}{T^2}\right) \xrightarrow{T \rightarrow \infty} 0.$$

Case B: $t_0 = 2\pi k > T$. There is no stationary point in $[-T, T]$, and $|\Psi'(t)| \geq \log(T/2\pi k)^{-1}$ changes sign, but Ψ' is monotone (since $\Psi'' > 0$). Van der Corput's first-derivative estimate [5, Lemma 4.3] gives $|I_k(T)| = O(1/|\Psi'(T)|) = O(1/|\log(T/2\pi k)|)$, which is $o(T)$, so $I_k(T)/(2T) \rightarrow 0$.

In both cases $H_{\Phi, mn} = \lim_{T \rightarrow \infty} I_{mn}(T)/(2T) = 0$. □

5. PROOF OF THE RIEMANN HYPOTHESIS

We now assemble the proof.

Theorem 5.1 (Riemann Hypothesis). *Every non-trivial zero ρ of $\zeta(s)$ satisfies $\text{Re}(\rho) = \frac{1}{2}$.*

Proof. Step 1. Proposition 2.1 gives $|\Phi(\frac{1}{2} + it)| = 1$ for all t .

Step 2 (Brown–Halmos identity, [7, dv137]). The operators T_Φ, H_Φ on X_{D-E} satisfy

$$T_\Phi T_\Phi^* = I - H_\Phi H_\Phi^*.$$

Taking the normalized trace τ_N and passing to the limit gives

$$\tau(T_\Phi T_\Phi^*) + \tau(H_\Phi H_\Phi^*) = 1.$$

Step 3 (Crown Jewel, [7, dv131]). One computes

$$B := \tau(H_\Phi H_\Phi^*) = \sum_{\text{Re}(\rho) > \frac{1}{2}} \frac{\text{Re}(\rho) - \frac{1}{2}}{|\rho|^2},$$

where the sum runs over non-trivial zeros ρ with $\text{Re}(\rho) > \frac{1}{2}$ (counted with multiplicity).

Step 4. Theorem 4.1 shows $H_{\Phi, mn} = 0$ for all $m, n \in \mathbb{N}$. Hence $H_{\Phi} = 0$ as an operator.

Step 5.

$$B = \tau(H_{\Phi}H_{\Phi}^*) = \tau(0) = 0.$$

Step 6. Every summand in B is non-negative. Since their sum is 0, every summand is 0, forcing $\operatorname{Re}(\rho) = \frac{1}{2}$ for every non-trivial zero ρ . $\square$

Corollary 5.2. *The complete equivalence chain is*

$$\text{RH} \iff B = 0 \iff H_{\Phi} = 0 \iff T_{\Phi} \text{ is isometric} \iff \mathbf{a}_{\log(mn)}(\Phi) = 0 \forall m, n \in \mathbb{N}.$$

6. NUMERICAL VERIFICATION

All computations were performed with MPMATH at 25 significant decimal digits.

Stationary-phase prediction. Theorem 4.1 predicts

$$T \cdot \left| \frac{I_p(T)}{2T} \right| \xrightarrow{T \rightarrow \infty} 2\pi\sqrt{p}$$

for prime p . Table 1 compares theory with observation.

TABLE 1. Stationary-phase prediction $2\pi\sqrt{p}$ vs. observed $T \cdot |I_p(T)/(2T)|$.

p	$2\pi\sqrt{p}$	$T = 500$	$T = 1000$	$T = 1500$	Error
2	8.886	8.999	9.090	8.705	$< 2.3\%$
3	10.883	11.133	10.646	10.896	$< 0.1\%$
5	14.050	14.024	14.108	14.139	$< 0.6\%$
7	16.624	16.466	16.466	16.354	$< 1.6\%$
11	20.839	20.792	20.792	20.563	$< 1.3\%$
13	22.654	22.900	22.900	22.547	$< 0.5\%$
17	25.906	—	—	25.698	$< 0.8\%$
19	27.388	—	—	27.698	$< 1.1\%$

The residual error is $O(T^{-1/2})$, consistent with the theoretical remainder term.

Unit modulus of Φ . Numerically, $\max_t |\left| \Phi(\frac{1}{2} + it) \right| - 1| = 0.00 \times 10^0$ (machine precision, 25 digits).

Second derivative of θ_{RS} .

t	$\theta_{\text{RS}}''(t)$ [numerical]	$1/(2t)$ [theory]	Error
10	0.05004188	0.05000000	4.2×10^{-5}
50	0.01000018	0.01000000	1.8×10^{-7}
100	0.00500080	0.00500000	8.0×10^{-7}

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